

Ruturaj Tambe

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Software Engineer experienced in performance engineering, distributed systems, & cloud platforms. Proven track record in scaling services, optimizing resource utilization, and reducing infrastructure costs. Currently expanding into ML-driven infrastructure optimization, seeking an opportunity to contribute to Cloud and AI infrastructure teams.

WORK EXPERIENCE

Avaya Software Engineer

R&D - CCAAS Analytics Team

Feb 2023 - May 2024

- Optimized Kubernetes resource allocation (**requests/limits, HPA tuning**), reducing OOM kills, throttling and stabilizing CPU/memory across **35+ critical issues**; instrumented Datadog & Splunk dashboards to inform scaling and SLOs.
- Led site performance and capacity engineering for **22 microservices**, improving throughput by **375%** and validating performance against SLOs (p95/p99) under peak load.
- Maintained automated load and capacity frameworks using Go, Python, and Locust, simulating **100K+ chat** and **20K+ voice transactions/hour** to analyze scalability and utilization.
- Built cost-aware capacity planning models to optimize headroom across **\$5M+ lab infrastructure**; an Azure migration plus Datadog log tuning delivered **\$4.2K/month** and **\$64.2K** in first-month savings while meeting reliability targets.
- Resolved 70+ system bottlenecks (Postgres lag, Redis imbalance), improving system resilience & fault tolerance.

Zensoft Services Pvt. Ltd. - Performance Engineering Team Lead

POC and Performance

Jul 2019 - Feb 2023

- Led performance engineering and capacity optimization for 20+ enterprise PwC projects; improved response times by **up to 500%** and scaled user concurrency **10x**
- Automated Azure DevOps & BlazeMeter pipelines reducing deployment time by **76% & 98%** using YAML workflow.
- Optimized client-side performance by reducing latency by **90%** through strategic caching and file-size reduction.
- Led **workload & capacity optimization** to enhance infrastructure utilization by **30%** & prevent scaling bottlenecks.
- Mentored & scaled the team from **5 to 23 engineers**, reducing onboarding time by **33%** and tripling delivery throughput.

New York University - Research Assistant

New York, USA

Stern School of Business | Wilf Family Department of Politics

Jan 2025 - Present

- **Scheduling Systems / Legin**: Developed **lekinpy**, a Python-based scheduling & optimization library implementing 4 classical job-shop scheduling algorithms; designed modular architecture for custom optimization strategies.
- **Political Economy of Corporate Fraud**: Developing a statistical modelling pipeline on 88,516 financial entities to estimate fraud probability using feature engineering and probabilistic modeling.

SKILLS

Programming : Python, Go, Bash, MySQL | **Platforms**: Linux, Windows, macOS

Cloud & Infra: Git, GCP, Azure, Kubernetes, Docker, Terraform, YAML, Jenkins

Performance Engineering: JMeter, Query Profiling using Datadog, Splunk, Postgres, Redis, Kafka, Prometheus, Grafana

Core Competencies: Capacity Modeling, Resource Allocation, Performance Benchmarking, Workload Simulation, Networking, Infrastructure Cost Optimization, Cross-functional Collaboration, Proof-of-Concept Ownership

EDUCATION

New York University Courant School of Mathematics & Stern School of Business

New York, USA

Master of Science in Information Systems

2026

Relevant Courses: Algorithms, Distributed Systems, Cloud & ML, Efficient AI & Hardware Accelerators, Deep Learning.

MIT, Savitribai Phule Pune University

Bachelor of Engineering in Computer Science and Engineering

Aug 2015 - June 2019

ACADEMIC PROJECTS

- **Infrastructure Benchmarking & Modeling (GPU/CPU)**: Micro-benchmark suite to compare DL training performance across environments (Mac M-series, CPU, and NVIDIA GPUs) on Tiny-ImageNet workload using roofline modeling.
- **Containerized AI Inference Service (Docker)**: Containerized a training workflow, systematically varying epochs, batch size, & learning rate to analyze their impact on model accuracy and execution time, documenting performance trade-offs
- **Neural Network Pruning & Efficiency Analysis (PyTorch)**: Implemented magnitude-based and structured pruning in PyTorch to study sparsity-accuracy tradeoffs and theoretical reductions in model compute (MACs) for efficient inference.
- **Automated GPU Provisioning (Terraform + GCP)**: Workflow to scan regions/zones, & auto-provision a T4-backed GPU VM.